Portaldot Network

—— Connecting People, Digital Assets,

and Future Intelligence

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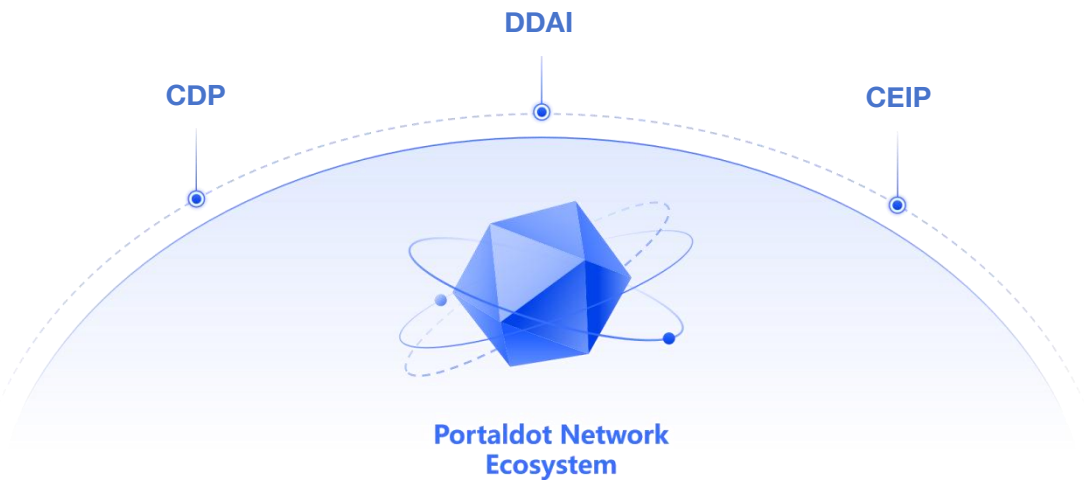
1. Portaldot Network: Connecting People, Digital Assets, and Future Intelligence

At the intersection of the Web3.0 wave and the Fourth Industrial Revolution, Portaldot Network is committed to building an open, secure, and sustainable blockchain ecosystem under the core vision of "connecting people, digital assets, and future intelligence." and promoting it to become the world's first full-dimensional Web3.0 infrastructure.Through three pillars—decentralized digital asset infrastructure,community driven platforms, and cutting edge integration platforms—we aim to deeply integrate the real and digital worlds, redefining the underlying logic of asset circulation, social collaboration, and technological innovation.



- **Decentralized Digital Asset Infrastructure**: Built on Portaldot's foundational technology, this infrastructure comprises four core modules:RWA Standardization Platform,iSwap DEX,iBridge Cross Chain Bridge, and ZKP Based Privacy Suite,forming a "highway" for seamless circulation and value exchange of real world assets, crypto assets, and data.
- **Community Driven Platform**: Portaldot's community platform operates under the principles of "co-creation, co-governance, and sharing," fostering open collaboration through volunteer and technical communities.
- **Cutting Edge Integration Platform**: Focused on AI large scale models, metaverse, and quantum computing, this platform drives synergistic innovation between blockchain and emerging technologies.

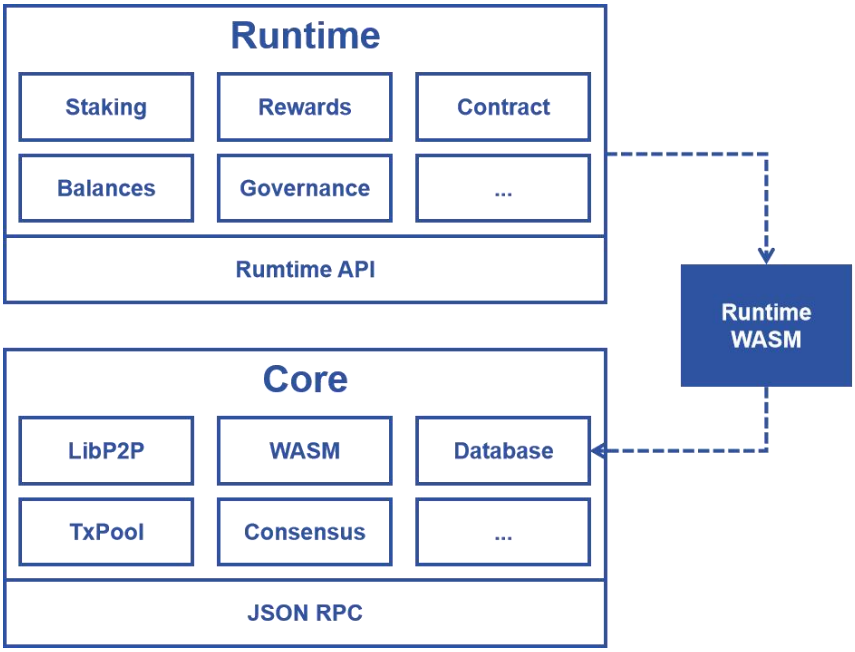
Portaldot Network is not merely a blockchain project but a social experiment in value networks. By integrating decentralized asset infrastructure, community governance, and cutting edge technologies,

we believe in achieving more efficient, equitable, and sustainable value exchange.

2. Portaldot Network Design

2.1 Architectural Design

How to design and implement an efficient and flexible blockchain system is a challenge we and many developers face together. As an innovative blockchain system, Portaldot Network divides the entire system into two core components, namely the Core part and the Runtime part, which together realize the synchronization, operation and final consensus of on-chain functions and on-chain data. By separating the basic functions and on-chain functions and running them synchronously, Portaldot Network not only ensures the stability and security of the system, but also improves the flexibility and scalability of on-chain functions.



2.1.1 Core Design

The Core is the cornerstone of Portaldot Network, which is mainly responsible for realizing the basic functions of a blockchain system. These functions constitute the core framework of the blockchain system and support the stable operation of the entire system.

- **Consensus system**: The consensus system is one of the core components of the blockchain system. It ensures that all nodes have consistent recognition of transactions and block data in a distributed environment. We use advanced consensus algorithms to achieve efficient data synchronization and verification while ensuring security. This design enables Portaldot Network to

maintain a high degree of stability and scalability when processing a large number of transactions.

- **P2P (Peer-to-Peer) communication**: P2P communication is the key technology for blockchain systems to achieve data exchange between nodes. By building an efficient P2P network, we achieve low-latency and high-reliability data transmission between nodes. This feature is crucial to ensure the real-time and availability of blockchain systems.
- **Data storage**: Data storage is a key link in blockchain systems to store transaction and block data. Portaldot Network uses Key-Value data storage technology and improves data redundancy and fault tolerance. At the same time, by optimizing the data storage structure, we have achieved efficient data retrieval and query functions.
- **Transaction pool**: The transaction pool is an area in the blockchain system that temporarily stores pending transactions. When designing the transaction pool, we ensured that it is highly flexible and scalable, capable of accommodating a large number of transactions and effectively sorting and verifying them. This design ensures that transactions can be processed and packaged into blocks in a timely manner according to priorities and rules.
- **Virtual Machine**: Virtual Machine is a key component of blockchain system to execute smart contracts and transactions. Portaldot Network's virtual machine uses an efficient and secure execution engine. This design enables developers to easily deploy and run smart contracts on the chain, thereby developing a variety of on-chain applications.
- **RPC interface**: The RPC interface is the window for the blockchain system to provide services to the outside world. By providing a rich set of RPC interfaces, external applications can easily interact with the blockchain system. These interfaces cover a variety of functions such as querying transaction information, submitting transactions, and obtaining block data, meeting the diverse needs of developers for the blockchain system.

2.1.2 Runtime Design

The Runtime part is another important part of Portaldot Network, which is mainly responsible for realizing on-chain functions. By compiling the Runtime part into runtime code and then synchronizing it to the Core part, we achieve efficient collaboration between the two.

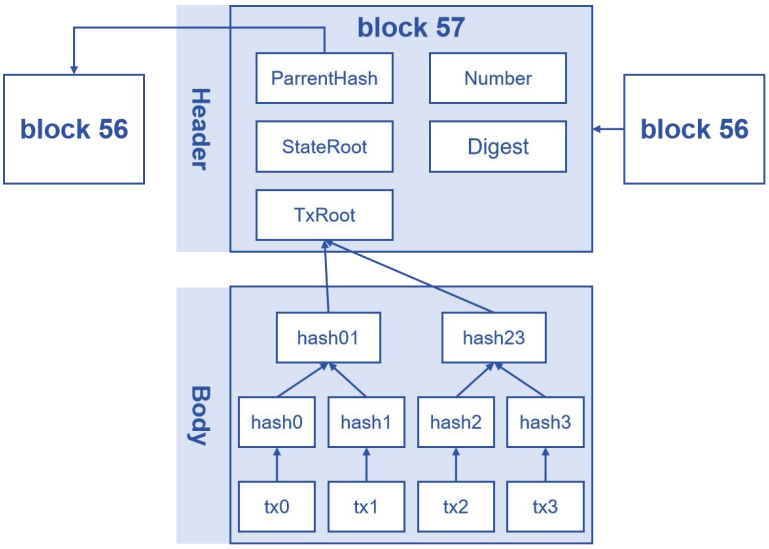
- **Staking system**: The staking system is a key link in our realization of decentralized governance and incentive mechanisms. Users can participate in on-chain governance and receive rewards by

staking tokens. This design not only ensures the fairness and transparency of the blockchain system, but also stimulates the enthusiasm of users to participate, providing a continuous source of power for the long-term development of the chain and ecology.

- **Incentive model**: The incentive model is an important mechanism to encourage user participation and contribution. The model will give corresponding rewards based on the user's behavior and contribution on the chain. This design not only encourages the active participation of users, but also promotes the prosperity and development of the chain ecosystem.
- **Account system**: The account system is the basis of user identity and assets. By building a secure and efficient account system, Portaldot Network ensures the uniqueness of user identity and the security of assets.
- **Smart Contract**: Smart contract is the key technology for blockchain system to realize automated execution and trustless transactions, which enables developers to easily build various on-chain applications. These applications cover multiple fields such as finance, games, and social networking, injecting strong vitality into our ecological development.

2.2 Block Structure

2.2.1 Block Header

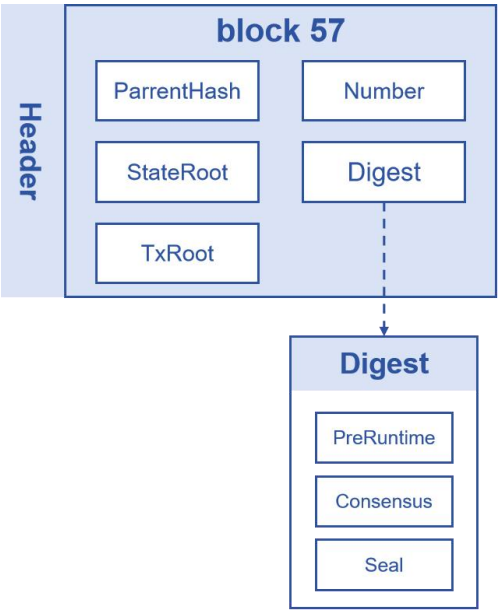


A block is a data structure that records transactions and status results that occurred over a period of time. It is a consensus on the current state of the ledger. A block is mainly composed of a block header and a block body. The block header consists of five parts:

- **ParrentHash**: parent block hash. Except for the genesis block, each block will point to its

own parent block, thus forming a chain structure of the blockchain;

- **Number**: the block height of the current block;
- **StateRoot**: State root, which is the proof of state changes after block execution. For details about state changes, see below;
- **TxRoot**: Transaction root, which represents the content and order of transactions in the block and is a summary of the block transaction information;
- **Digest**: additional information of the block.



In Portaldot Network, the Digest part mainly contains the proof information of the block consensus to ensure that this is a legal block, which includes three parts:

- **PreRuntime**: Proof of block producer, such as VRF (Verifiable Random Function) proof, etc.
- **Consensus**: consensus algorithm information, such as validator switching information;
- **Seal**: The signature of the validator who produced the block, indicating who produced the block.

2.2.2 Block Body

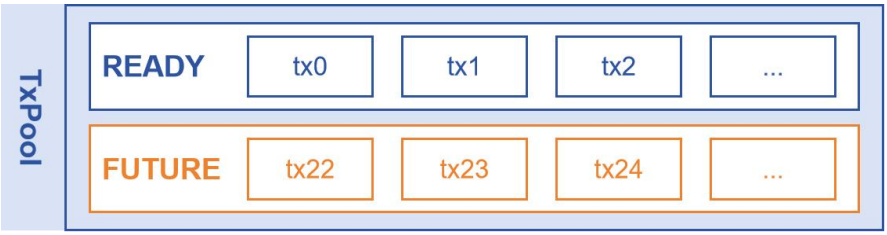
The block body consists of all transactions contained in the block. The root hash obtained by organizing these transactions through the MPT (Merkle Patricia Tree) tree corresponds to the TxRoot in the block header.

In addition, each block contains a special transaction to mark the generation time of the block, which can also avoid the generation of empty blocks.

2.3 Transaction Pool and Transactions

2.3.1 Transaction Pool

The transaction pool contains all transactions broadcasted in the blockchain network and received and verified by the local node. These transactions may be signed or unsigned.



The work of the trading pool includes:

- **Validity check**: Check whether the transaction is valid, otherwise it will not be allowed to enter the transaction pool. The transaction pool also regularly checks the validity of existing transactions in the pool. If an invalid or expired ordinary transaction is found, the transaction will be deleted by the transaction pool.
- **Transaction sorting**: If the transaction is valid, the transaction queue will divide the transaction into two groups:

  1) Ready Queue - contains all transactions that can be put into a new pending block.

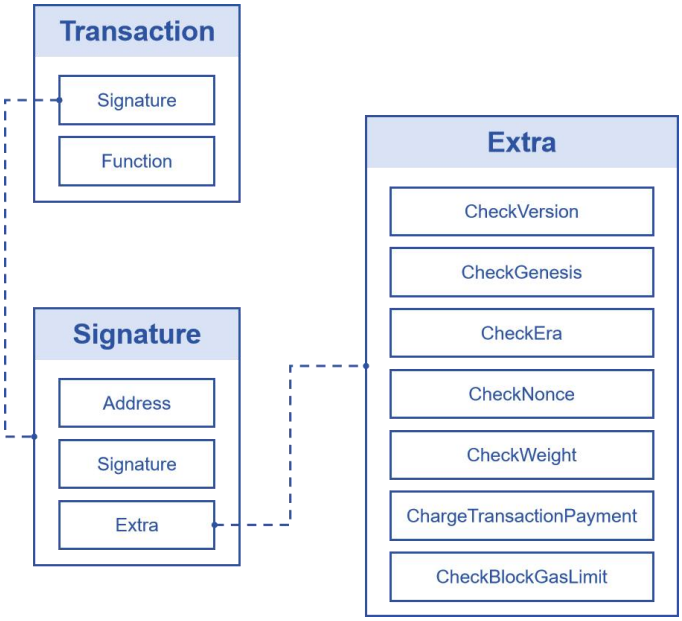
  2) Future Queue - contains all transactions that may become valid in the future.。
- **Dependency Management**: Build transaction dependencies to allow the transaction pool to generate a valid linear order of transactions, even if all transactions from a single sender will form a valid order in the pool.
- **Determine transaction priority**: When a node becomes the next block generator, it will sort the transactions in that block from high to low priority until the block weight or fee cap is reached.
- **Enter the transaction life cycle**: responsible for the execution of the transaction and delete the transaction from the transaction pool when the block is imported.

2.3.2 Transaction Structure

In Portaldot Network, all on-chain operations are completed by issuing and executing a transaction. Each transaction consists of two parts:

- **Signature**:Validates sender and integrity;

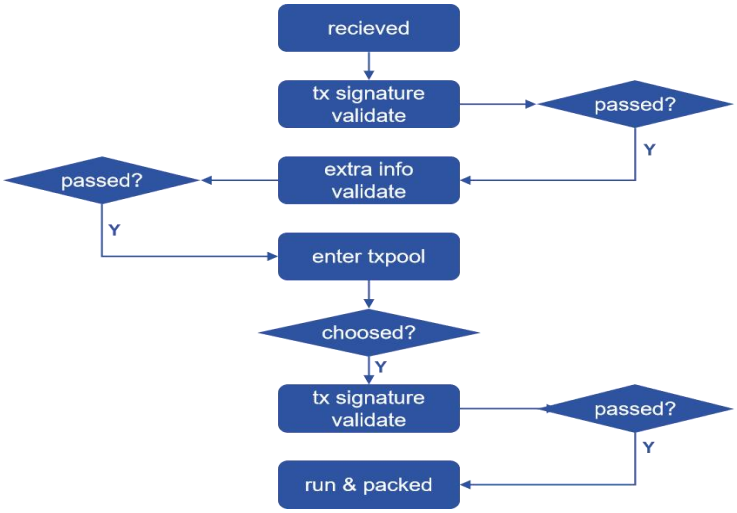
- **Function**: Defines transaction purpose (e.g., transfers, contract calls).



The transaction signature is further divided into three important components:

- **Address**: allows the entire network to mark the initiator of the transaction;
- **Signature**:Aims to confirm the authenticity and integrity of the transaction and ensure that the transaction has not been maliciously tampered with during the transmission process;
- **Extra**:This part covers multiple data verifications. For example, version information enables the network to accurately identify the rule version followed by the transaction. The survival time stipulates the time range in which the transaction is valid in the transaction pool. The transaction fee is the fee paid by the user for using the network service. The transaction weight reflects the proportion of resources in a block.

2.3.3 Transaction Execution Lifecycle

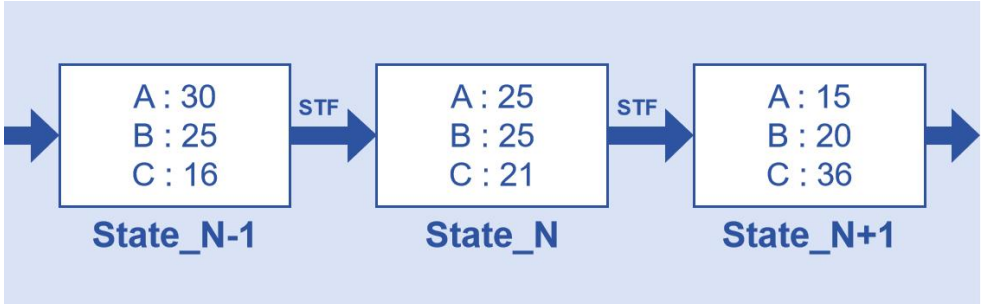


In Portaldot Network, a transaction execution consists of multiple steps:

- **The node receives the transaction request sent by the user;**
- **Verify the legitimacy of the transaction signature;**
- **Extra info validate;**
- **Enter txpool;**
- **Whether selected to the ready queue or not;**
- **Re-verify the transaction signature;**
- **Run & Packed.**

2.4 Chain State and MPT Based Implementation

2.4.1 STF Overview



STF（State Transition Function）is a concept widely used in many fields such as computer science, automata theory, control theory, etc. It describes how a system transitions from one state to another based on the current state and input.

Generally, an STF can be expressed in the following mathematical form:

$$\delta : Q \times \Sigma \rightarrow Q$$

$$Q : \ the\ set\ of\ all\ possible\ states$$

$$\Sigma : \ the\ set\ of\ all\ possible\ inputs\ or\ events$$

$$\delta(q, \sigma) \ = \sigma'$$

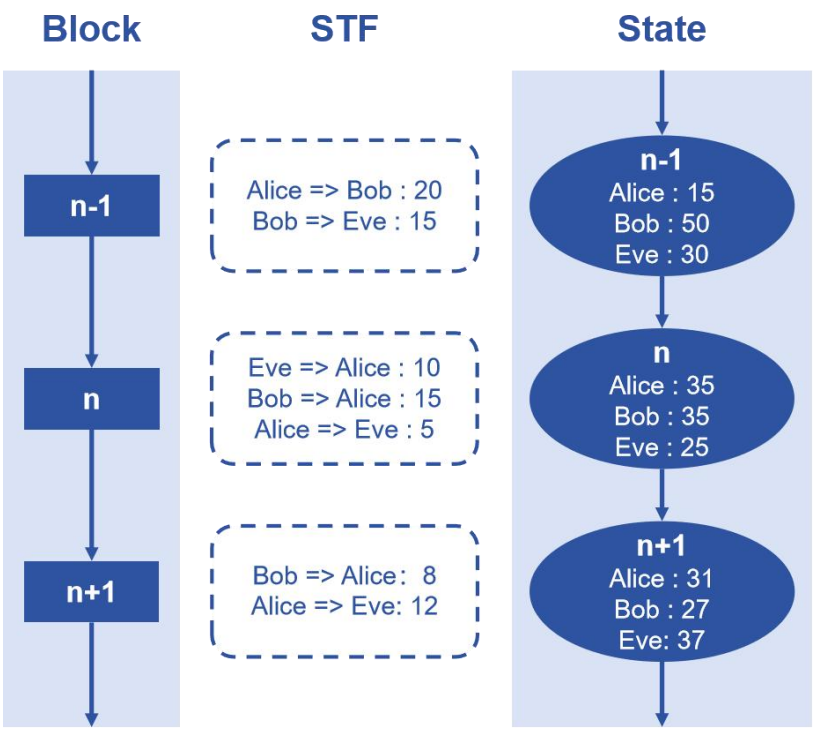
$$q \in Q, \ \sigma\ and\ \sigma' \in \Sigma$$

2.4.2 From STF to Blockchain

The current mainstream blockchain systems mainly use two types of models when modeling business:

- **UTXO**,a model represented by Bitcoin and its branches;
- **State Model**,A model that focuses on recording status, represented by Ethereum.

Considering that the state model can provide better support in supporting a wider range of scenarios, more blockchain systems will still choose the state model when modeling.



In a state model blockchain system, STF mainly deals with two contents:

- **Transaction processing**: Blockchain is essentially a distributed ledger that records a series of transactions, and the state transfer function updates the global state of the blockchain based on new transactions.
- **Smart contract execution**: A smart contract is an automatically executable code on the blockchain. When an external request triggers a smart contract, the state transfer function executes the contract code and updates the state of the blockchain based on the execution results.

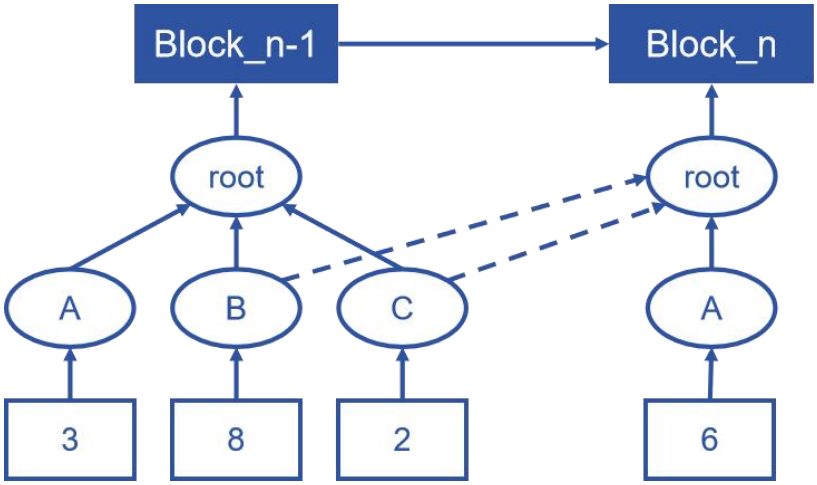
For a state model blockchain system, STF needs to have the following characteristics:

- The system state under the latest block is the current state, and the state of all current objects can be obtained in this state;
- The block contains proof of the state of the block, which means that the current state has been agreed upon by the nodes.
- You can obtain the status under any historical block, rather than the status under the latest block.

2.4.3 MPT Based State Transition

In order to meet the above conditions, Portaldot Network adopts an MPT-based algorithm in the process of managing state migration, making the process of tracking state modifications highly efficient and significantly saving resources.

Whenever a state change occurs, we do not re-record the entire state, but only record the data that has been modified due to the state change and form a new MPT root. This root is called state_root in the block header and serves as the state proof for the consensus of the current block. On the other hand, for those parts of the state that have not changed, the system directly references the index of the previous state, which not only maintains the integrity of the data, but also saves a lot of storage space and computing resources.



As shown in the figure above, in Block_n-1, the process from root > A > 3 is equivalent to recording the value of A as 3, and the process from root > B > 8 is equivalent to recording the value of B as 8. In the process from block_n-1 to block_n, the state of A changes from 3 to 6. At this time:

- Since the value of A has changed, MPT generates a new leaf node to represent the new state of A, and then regenerates a new index path from the leaf node, that is, the process of 6 > A > root in block_n;
- Since the status of B and C has not changed, when generating a new path, the paths of B and C in blcok_n-1 are directly referenced.

There are many advantages to using an MPT-based algorithm to record state transitions in Portaldot Network.

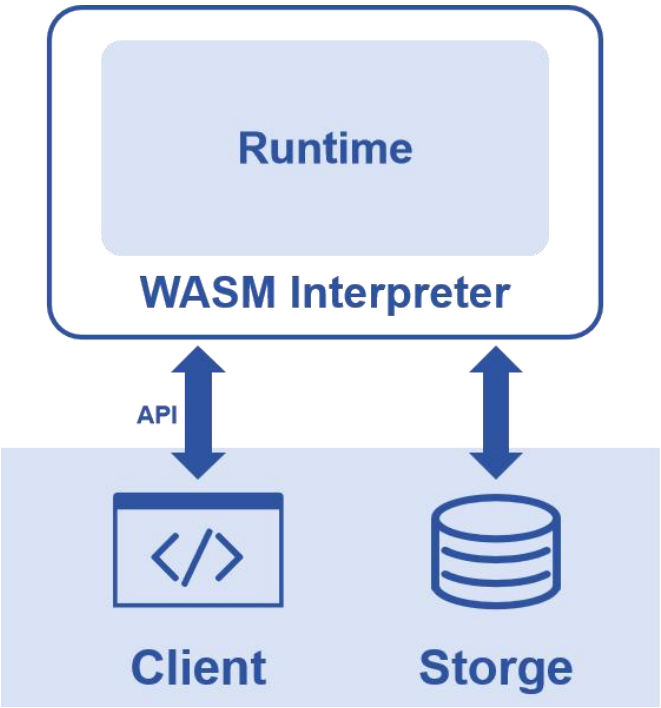
- Improved the speed of state migration. Since only modified data is processed and included in

the state root, the time required to update the state is significantly reduced;

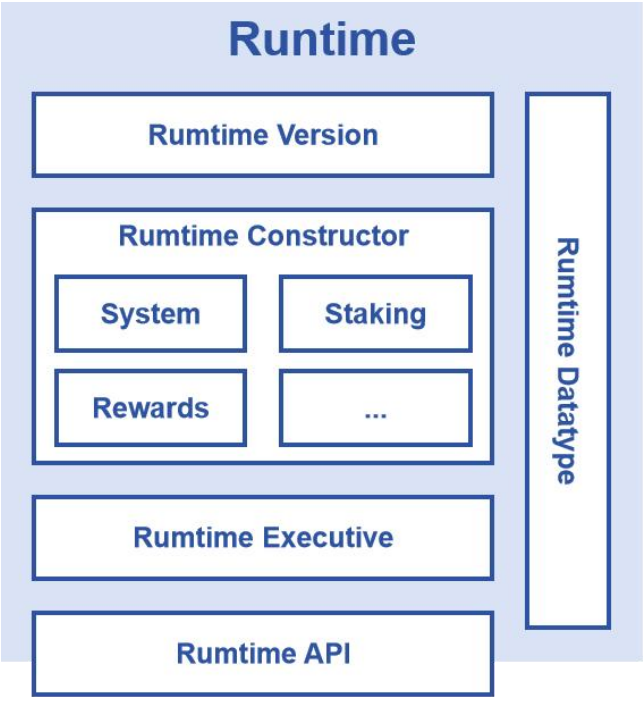
- Improved storage efficiency. By minimizing the amount of redundant data stored, the network can manage more state information with fewer resources;
- Because MTP ensures that any changes to the state can be verified, it also helps improve the overall security and integrity of the network.

2.5 Runtime and Hot Updates

2.5.1 Runtime Structure



For traditional applications, programs are divided into two parts: instructions and data, and need to run in a suitable operating system. In order to understand the role of Runtime in Portaldot Network more vividly, we can regard the on-chain code Runtime as program instructions, the on-chain storage Runtime Storge as program data, and the WebAssembly Interpreter as the operating system. Once this process is completed, the entire system can be put into operation to provide services to users. In addition, it should be noted that the Runtime we are going to discuss here only includes the on-chain code part.



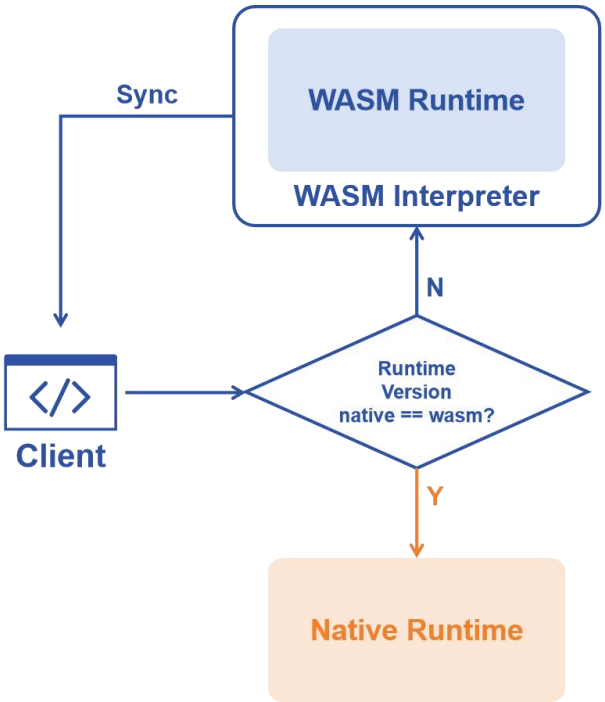
Runtime consists of five parts:

- **Version**: Used to mark the current Runtime version. Every update and improvement needs to be clearly presented to the nodes in the entire blockchain network through version control.
- **Constructor**: A macro used to construct all functions of the Runtime, laying the foundation for the normal operation of the Runtime. A Runtime contains many complex functional modules, which work together to realize various functions of the blockchain, such as block generation and verification, transaction processing and execution, etc.
- **Executive**: It is the Runtime executor, which provides a full range of interfaces for the normal operation of the Runtime. These interfaces cover a series of key operations in the operation of the blockchain system, such as initialization, block execution, transaction execution, and transaction verification.
- **Datatype**: A "dictionary" of chain-related type definitions, mainly responsible for some chain-related type definitions, such as blocks, transactions, accounts, etc., providing clear definitions and specifications for various data elements in the blockchain system.
- **API**: The interface for Runtime to interact with the outside world. For developers, API provides them with a convenient way to call the functions of the blockchain system.

The five parts of Runtime do not exist in isolation, but work together to form an organic whole. Version provides version information for the entire system to ensure that the various parts can interact

and work together correctly in different versions of Runtime. The modules built by Constructor provide the Executive with the basic functions required to perform operations, and the Executive calls these modules to realize the core operation control of the blockchain system, including the processing and verification of blocks and transactions. The data type defined by Datatype is the basis for data processing and interaction of the entire Runtime. Whether it is the Constructor building modules or the Executive performing operations, it depends on the standard data structure defined by Datatype. API serves as a bridge for the entire Runtime to communicate with the outside world, providing internal functions and data to the outside world in the form of standardized interfaces, and also introducing external requests into the Runtime for processing.

2.5.2 Runtime Hot Updates



In Portaldot Network, there are two types of Runtime. One is the Runtime code included in the client during compilation, which we call Native Runtime. The other is the Runtime code deployed to the chain, called WASM Runtime, which is available to everyone. When executing the Runtime function, the client will check whether the two versions are consistent. If they are consistent, the client will delegate the execution of the Runtime function to Native Runtime. When WASM Runtime is updated, if the client has not been updated, the node will execute the latest Runtime on the WASM Interpreter virtual machine integrated in the client.

Therefore, under any circumstances, all nodes on the network can always correctly synchronize the

chain and implement hot updates of the on-chain code, thereby achieving a fork-free upgrade of the chain.

2.6 Key Management System

In Portaldot Network, we use different account types to distinguish different permissions and functions. We roughly divide them into account keys that users interact with and session keys that nodes manage without operator intervention.

2.6.1 Account Keys

Account keys are associated with a certain balance, and parts of these balances can be locked to play a role in staking, resource leasing, and governance, including several different types of unlocking periods. We allow multiple locks with different periods, partly because these roles have different restrictions, and partly because multiple unlocking periods can be carried out at the same time.

We encourage users to actively participate in all on-chain roles, but all of these roles require accounts to sign operations. At the same time, for better security, some users may store account keys in places that are not easy to access, but this also makes signing operations cumbersome, so we avoid this inconvenience in this way: accounts that lock funds for staking are called "reserve accounts". All reserve accounts will register a certificate on the chain. This certificate delegates all validator operation permissions and nomination permissions to a control account, and also specifies a proxy key for voting. In this state, the control account and the proxy account can sign for the reserve account in the staking and governance functions respectively, but cannot transfer funds.

Currently, we support ed25519 and Schnorrkel/sr25519 as account keys. Both are Schnorr-like signature algorithms implemented using the Ed25519 curve, so they provide very similar security. For users who need HSM (Hardware Security Module) support or other external key management solutions, we recommend using ed25519 keys, while Schnorrkel/sr25519 provides more convenient features such as HDKD (Hierarchical Deterministic Key Derivation) and multi-signature.

In particular, Schnorrkel/sr25519 uses Mike Hamburg's Decaf's Ristretto implementation, which treats the Ed25519 curve's 2nd-free torsion points as a prime order group. Avoiding cofactors like this means that Ristretto makes it safer to implement more complex protocols. We use the Blake2b algorithm for most general hashing operations, but Schnorrkel/sr25519 itself uses the STROBE128 algorithm, which is based on Keccak-f(1600) and provides a hashing interface that is well suited for

signatures and NIZKs (Non-Interactive Zero-Knowledge Proofs).

2.6.2 Session Keys

Session keys have a specific role in consensus or security. In general, session keys are only authorized from a session certificate signed by a control key that delegates the corresponding stake.

At any time, the control key can suspend or revoke the session certificate and issue a replacement certificate with a new session key. All new session keys can be pre-registered, and in most cases must be pre-registered, so that validators can smoothly transition to new hardware by issuing session certificates that are only valid after a future session begins. We recommend using the suspension mechanism for emergency maintenance and the revocation mechanism if a session key may have been compromised. However, we prefer to tie session keys to a physical machine because this minimizes the risk of accidentally creating inconsistencies. We require validator operators to use the RPC protocol to issue session certificates.

When validators produce blocks, they use Schnorrkel/sr25519 keys for regular Schnorr signatures, and also for NSEC5-based VRF, which verifies that the block was legitimately produced. When the block determines the final consistency, the validator uses BLS (Boneh Lynn Shacham) signatures to vote. BLS signatures support convenient signature aggregation. For performance considerations, we chose the BLS12-381 curve.

2.7 Smart Contracts

Portaldot Network provides the ability to deploy and execute WebAssembly-based smart contracts. We recommend using the ink! language for smart contract development.

2.7.1 Ink!

ink! is an eDSL (Embedded Domain-Specific Language) based on Rust, which is dedicated to writing WASM smart contracts in blockchain networks. Its design aims at correctness, simplicity, and efficiency, and has many advantages:

1） Based on Rust

- **High security**: Rust has strong memory safety and concurrency safety features. It can capture many common memory errors, such as null pointer references and data competition, at the compilation stage through mechanisms such as ownership system and borrow checker, which greatly reduces potential security vulnerabilities in smart

contracts.

- **Superior performance**: Rust is a system-level programming language with performance close to that of C and C++. ink! inherits the high performance of Rust, making smart contracts written based on ink! perform well in execution speed and resource utilization efficiency, and can handle high-concurrency transaction requests.
- **Rich ecosystem**: Rust has a huge developer community and rich third-party libraries. Ink! Developers can use these resources to quickly implement various functions and improve development efficiency.

2） Simplicity and efficiency

- **Simple code**: As an eDSL, ink! provides simple syntax and abstraction, allowing developers to implement complex smart contract logic with less code. This reduces development costs and maintenance difficulties, while also improving code readability and maintainability.
- **Efficient development**: ink! provides a series of macros and tools to help developers quickly build the framework of smart contracts. Developers can focus on the implementation of business logic without paying too much attention to the underlying details, thereby improving development efficiency.

3） Correctness guarantee

- **Compile-time checking**: ink! uses Rust's compile-time checking mechanism to detect many potential errors during the compilation phase and ensure the correctness of smart contracts. This greatly reduces the possibility of errors at runtime and improves the reliability of smart contracts.
- **Formal Verification Support**: ink! has some support for formal verification. Formal verification is a technique that uses mathematical methods to prove the correctness of a program, which can further improve the security and reliability of smart contracts.

2.7.2 Smart Contract Development

Traditional smart contract platforms allow users to publish additional logic on top of the core blockchain logic. Since anyone, including malicious attackers, can publish smart contracts, these platforms have many deliberately designed security protections built in. Include:

- **Transaction fees and state leasing**: Ensure that users pay for the computing and storage resources used to deploy or execute smart contracts, so that the resources of the block-producing nodes are not abused by them.
- **Sandbox mechanism**: A contract cannot directly modify the core blockchain storage or the storage of other contracts. Its role is to limit the contract to modify only its own state and to initiate external calls to other contracts.
- **Rollback**: The contract may have situations that lead to logical errors, and the rollback mechanism can prevent any state from being changed when an error occurs.

These settings make smart contracts run slower and more expensive, but the "target audience" of contract development is different from blockchain developers. Contracts allow the community to expand and develop on your application without having to deal with various underlying issues. Therefore, we have made a lot of optimizations to the smart contract module of Portaldot Network to ensure:

- The security of the system, error handling introduced at an earlier stage;
- Abuse can be prevented through financial incentives;
- More efficient developer accessories.

2.8 POT

POT is the native digital asset of Portaldot Network, which adopts the initial issuance + limited dynamic inflation reduction model. The initial issuance is 900,000,000. The inflation rate depends on the overall staking level of the network. It is halved every 4 years. The maximum inflation planned for 20 years is 2,100,000,000, and the final issuance is 3,000,000,000. Relying on the decentralized nature of the blockchain network, POT has a high degree of security, transparency and immutability, and its operation and transaction fairness are jointly maintained by nodes around the world. The role of POT in the entire ecosystem mainly includes:

- **Staking**: Staking is one of the core functions in the POT ecosystem. Holders can stake their POT in the network and participate in the maintenance of the consensus mechanism of the blockchain. Through staking, on the one hand, it helps to enhance the security and stability of the entire network, and the interests of the pledgers are closely linked to the healthy operation of the network; on the other hand, the pledgers can obtain corresponding rewards.

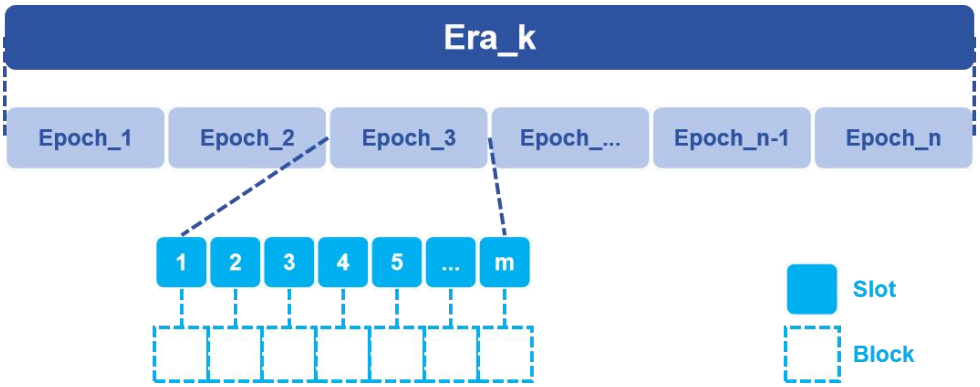
This reward mechanism encourages more users to participate in staking, thus forming a virtuous circle and continuously improving the overall staking rate and security of the network. For investors, staking POT provides a relatively stable and attractive way to obtain income. In an environment where the cryptocurrency market is volatile, staking income can be used as an important supplement to asset appreciation.

- **Payment**: POT acts as a medium of value circulation in the Portaldot Network blockchain network. Various operations in the network require the use of POT, including transfers, smart contracts, pledges, etc., which promotes the efficient exchange and flow of resources and value.
- **Ecological circulation:** In the process of developing ecological applications, developers can obtain POT as financial support and rewards for project development, which can motivate them to continue to innovate and optimize applications; when using ecological applications, users can use POT to pay service fees or obtain value-added services within the application; merchants can expand sales channels and customer groups by accepting POT payments. This wide range of ecological circulation gives POT an intrinsic value support. Its value does not only depend on market speculation, but is also closely related to the prosperity of the ecosystem. As the Portaldot Network ecosystem continues to grow and develop, the circulation demand for POT will continue to increase, thereby driving the steady increase in its value and bringing long-term investment return expectations to investors.

3. LAO NPoS: A New Consensus and Incentive Mechanism

3.1 Conceptual Framework

3.1.1 Time Settings

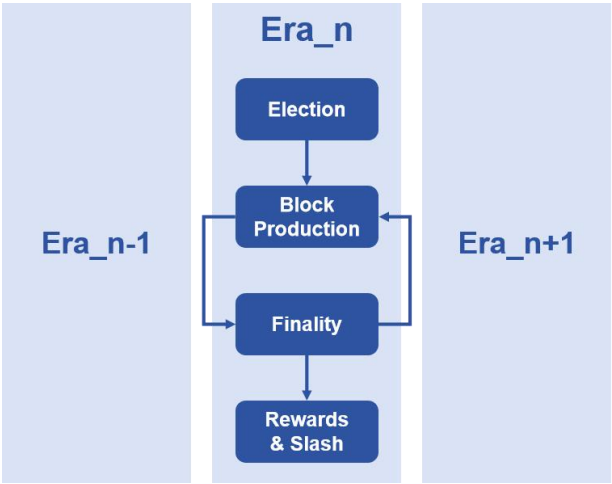


- **Slot**: Represents the block generation time. A block will be generated at the end of each slot.
- **Epoch**: Represents a set of slots of fixed length. The randomness parameters related to block production will be redefined in each period.
- **Era**: Represents a fixed-length Epoch set, and validators will be re-elected at the end of each epoch.

3.1.2 Role Definitions

- **Validator**: After each Era, the system selects a set of validators who will play a key role in highly sensitive protocols in the following Era, such as block production and final verification. Their work is demanding, as they need to run costly operations, ensure high communication responsiveness, and build a long-term reputation for reliability. They must also stake their tokens as a guarantee of good behavior, which will be punished when they deviate from the protocol. On the contrary, they are rewarded when they play by the rules. Any node that completes the task can publicly become a candidate for verification node. However, due to operational reasons, only a limited number of validators can be selected in the end.
- **Nominator**: The nominator publishes a list of candidates he trusts and stakes a certain amount of tokens to support them. If some of these candidates are selected as validators, he will receive corresponding benefits in proportion to the amount of staked. Unlike validators, there is no limit on the number of nominators. As long as the nominator chooses and only supports candidate validators with good security practices, his risk is low and there is a continuous source of income.

3.2 Consensus and Incentive Workflow



In Portaldot Network, we use a hybrid consensus instead of a traditional single consensus. The complete consensus process is divided into three stages:

- Phase 1: At the beginning of each Era, validator election is conducted;
- Phase 2: Validators produce blocks;
- Phase 3: final confirmation;
- Phase 4: At the end of each Era, incentives and penalties are calculated.

3.3 Validator Election

How should validators be selected when the nominator's voting status is known? Unlike other traditional PoS (Proof of Stake) projects, Portaldot Network does not only determine the weight based on the validator's shareholding, but also gives validators equal voting rights in the consensus protocol. To reflect this fact, the nominator's shareholding should be distributed to the elected validators as evenly as possible, while still respecting the nominator's preferences. Therefore, the validator election algorithm we designed follows two principles:

- **Fair representation**: ensures that any group of nodes is neither too high nor too low in representation among the elected validators, and is proportional to their stake. Assuming we need to elect k validators, any nominator holding at least 1/k of the total stake is guaranteed to have at least one of the validators they trust elected.
- **Safety**: If a nominator has two or more validators elected that they trust, we need to distribute the nominator's stake between those validators so that the support of the validators is as balanced as possible. Remember, we want to make it extremely difficult for an adversarial group of nodes to get a validator elected, and an adversarial group can only do this if they have high enough support. Therefore, we equate the safety level of the election result to the minimum support of any elected validator.

3.4 VRF Based Block Production

3.4.1 VRF Overview

VRF is a verifiable random number generation function, which mainly includes two types of functions: generation function and verification function.

$generating\ function$	$R = VRF_Hash(sk, M)$ $P = VRF_Proof(sk, M)$
$validating\ function$	$R = VRF_P2H(P)$ $VRF_Verify(pk, M, P)$

$$
\begin{aligned}
sk &: private\ key \\
pk &: public\ key \\
M &: \ input \\
R &: \ VRF\ output\ hash \\
P &: \ VRFproof
\end{aligned}
$$

In the application process, the prover first generates a pair of public and private keys, and then uses the generation function to generate a hash value R and a hash value proof P respectively, and the verifier uses the verification function to verify R and P respectively. Therefore, the VRF function has the following characteristics:

- **Randomness**: For different input values, the output value is random and evenly distributed within the value range;
- **Determinism**: For the same input value, the output value it obtains must be the same;
- **Verifiability:** Non-interactive zero-knowledge proof can be used to verify the authenticity of the random number output, indicating the legitimacy of the random number;
- **Efficiency**: Off-chain computing improves the efficiency of the process.

3.4.2 Block Production

Portaldot Network block production is anchored to Slots and uses the VRF PRNG method to randomly perform slot allocation. On each Slot, all selected validators will generate a random number using the VRF PRNG generation function. If the random number is less than a given threshold (proportional to their weight/stake), they have the right to produce a block. The process is as follows:

- The hash length (hashlen) of the output value is fixed, so converting the output hash to a number gives the range of the hash:

$$[0, 2^{hashlen} - 1]$$

- The node weight should be positively correlated with the probability of being selected:

$$W = \sum w_i$$
$$p = \frac{w_i}{W}$$

$$\begin{aligned} \omega_i &: \ weight\ of\ node\ i \\ W &: \ total\ weight \end{aligned}$$

- The probability of node w being selected conforms to the binomial distribution:

$$B(k; w, p) = C_w^k p^k (1-p)^{w-k}$$
$$\sum_{k=0}^{w} B(k; w, p) = 1$$

The binomial distribution is used as the probability distribution function to prevent Sybil attacks, because

$$B(k_1 + k_2; w_1 + w_2, p) = B(k_1; w_1, p) + B(k_2; w_2, p)$$

Therefore, from a probabilistic point of view, it is impossible to increase the possibility of being selected by splitting the token.

- Split the space [0, 1) into continuous intervals:

$$\left[\sum_{k=0}^{j} B(k; w, p), \sum_{k=0}^{j+1} B(k; w, p)\right)$$
$$j \in \{1 \ldots w\}$$

- If the ratio of the hash value to the maximum hash value falls within the interval j>0, it indicates that the node is selected. The overall algorithm of the election lottery is as follows:

$$\begin{aligned}
&Sortition(sk, seed, \tau, role, w, W):\\
&\langle hash, \pi \rangle \leftarrow \mathrm{VRF}_{sk}(seed \| role)\\
&p \leftarrow \frac{\tau}{W}\\
&j \leftarrow 0\\
&while\ \frac{hash}{2^{hasshlen}} \notin \left[\sum_{k=0}^{j} B(k; w, p), \sum_{k=0}^{j+1} B(k; w, p)\right)\ do\\
&\qquad j++\\
&return\ \langle hash, \pi, j \rangle
\end{aligned}$$

$$\begin{aligned} csk &: \ private\ key \\ seed &: \ PRNG\ seed \\ role &: \ role\ value \\ \tau &: \ threshold \end{aligned}$$

- Other peer nodes can use the proof generated by the VR PRNG method to verify the legitimacy of the block statement. The verification method is as follows:

$$
\begin{aligned}
&VerifySort(pk, hash, \pi, seed, \tau, role, w, W):\\
&if\ VerifyVRF_{pk}(hash, \pi, seed||role)\ then\ return 0;\\
&p \leftarrow \frac{\tau}{W}\\
&j \leftarrow 0\\
&while\ \frac{\text{hash}}{2^{\text{hashlen}}} \notin \left[\sum_{k=0}^{j} B(k; w, p), \sum_{k=0}^{j+1} B(k; w, p)\right) do\\
&\qquad j++\\
&return\ j
\end{aligned}
$$

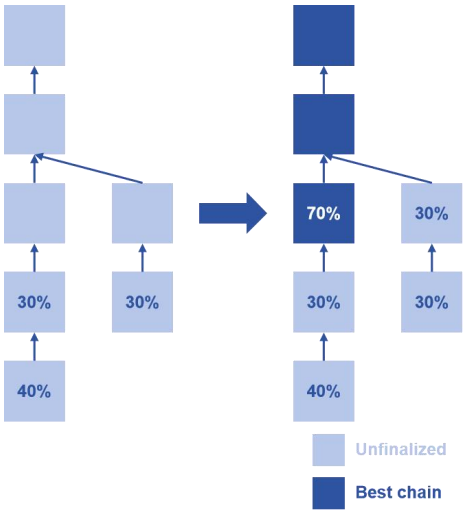
Within an Epoch, we will use the same validator set, and all VRF outputs generated by block generation will be collected in the on-chain random pool. At the end of Epoch number N, the Epoch parameters of N + 2, such as validator weight, VRF PRNG seed, etc., are announced on the chain.

Since slot allocation is random, it is possible that a slot is assigned to multiple validators, in which case we will have a temporary fork, or that a slot is not assigned to any validator, in which case no blocks will be produced. In this case, the block time is uncertain.

In addition to the above VRF-based slot allocation (which we call primary slot), we also support deterministic secondary slot allocation. When running a validator node, the primary slot has a higher priority than the secondary slot, and the validator node first tries to claim the primary slot and then falls back and tries to claim the secondary slot. Secondary slot allocation is done by selecting authority at the index.

3.5 BFT Based Finality

Based on the BFT (Byzantine Fault Tolerance) protocol, Portaldot Network has designed an eventual consistency algorithm for the longest chain. We assume that more than 2/3 of the voters vote honestly. The final block is determined through two rounds of voting (pre-voting and pre-committing). Participants vote for the chain they think is the best, thereby trying to finalize the largest confirmed block agreed by more than 2/3 of the participants. The algorithm can finalize multiple blocks at a time.

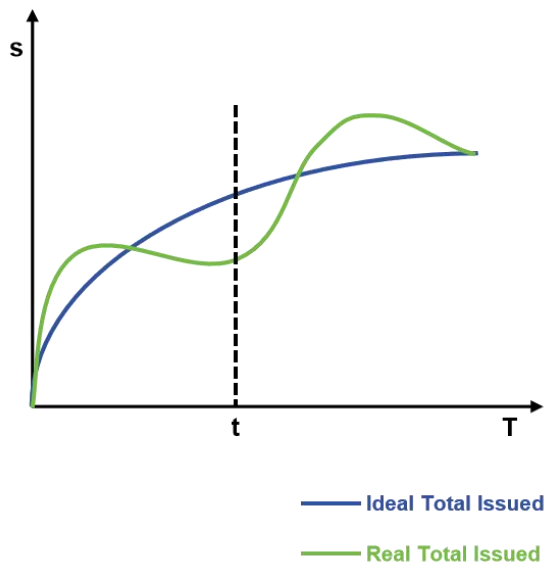


The algorithm works as follows:

- **Highest block broadcast**: The validator of the newly determined block broadcasts the final highest block hash of the previous round;
- **Pre-vote**: After waiting for network delays, each validator will broadcast a "pre-vote" for what it believes should be the final highest block that has been determined. In the case where the vast majority of validators are honest, the validator will broadcast the highest final block hash to confirm which candidate blocks are extended on the longest chain. This new chain may have more blocks than the last final chain that has been determined;
- **Pre-commit**: Each validator calculates the new highest block that can be finalized based on the pre-vote set. If the pre-vote set extends the last finalized chain, then each validator will "pre-commit" to that chain;
- **Commit**: Each validator waits to receive more than 2/3 of the pre-commits to form a commit message on the newly determined chain and broadcast it to other nodes.

3.6 LAO Algorithm Incentives

In Portaldot Network, the growth of POT mainly comes from the income generated after staking. With the continuous growth of consensus participants, the total issuance of POT has also begun to grow explosively. In order to prevent the inflation that may be caused by this growth and ensure the value growth of POT, we have introduced a new mechanism to the incentive layer of the consensus - the LAO (Linear Attenuation Offset) algorithm.



We assume that there is a curve, the ideal total amount curve, which represents the ideal value of money supply at a given point in time. The LAO algorithm focuses on the relationship between the actual money supply at time point t and the ideal curve. The goal is to calculate the adjustment strategy we need to make to money output after a given k Era ends after time point t. The mathematical process is expressed as follows:

$$\delta = s_{real}(t) - s_{ideal}(t)$$
$$O : \sum_{i=0}^{k-1} o_i = \delta \ , \quad o_i \in O$$
$$s(t) : \ issued\ amount\ at\ t$$
$$O : \ the\ set\ of\ offsets$$

In a traditional NPoS mechanism, the incentive output after each Era is determined by the annual inflation rate I, which is determined by the pledge rate x. When x reaches the ideal pledge rate, I reaches its maximum value. The mathematical process is expressed as follows:

$$I(x) = \begin{cases} I_0 + (I(x_{ideal}) - I_0)\frac{x}{x_{ideal}} \ , & 0 < x \leq x_{ideal} \\ I_0 + (I(x_{ideal}) - I_0) \times 2^{(x_{ideal} - x)/d} \ , & x_{ideal} < x \leq 1 \end{cases}$$
$$I_0 : \ the\ limit\ of\ I(x)\ as\ x = 0$$
$$d : \ decay\ rate$$

Portaldot Network has a design that reduces the Era incentive under the NPoS algorithm every four years. After the introduction of the LAO algorithm, assuming that the number of Eras included each year is z, then in the k Eras after time t, the incentive r for each is:

$$r = \beta \cdot \frac{I(x) \cdot s_{real}(t)}{z} + o_i \ , \quad 0 \leq i < k$$

$$\beta : \quad rewards\ decay\ rate$$

4. Portaldot Network Ecosystem Design and Roadmap

4.1 Decentralized Digital Asset Infrastructure: Trust in Asset Circulation

In the construction of the Portaldot Network ecosystem, creating a practical, convenient and secure decentralized digital asset base is one of the themes. This is not only a technical concept, but also a bridge connecting the digital world and the physical world, opening up a new path for future asset circulation and financing.

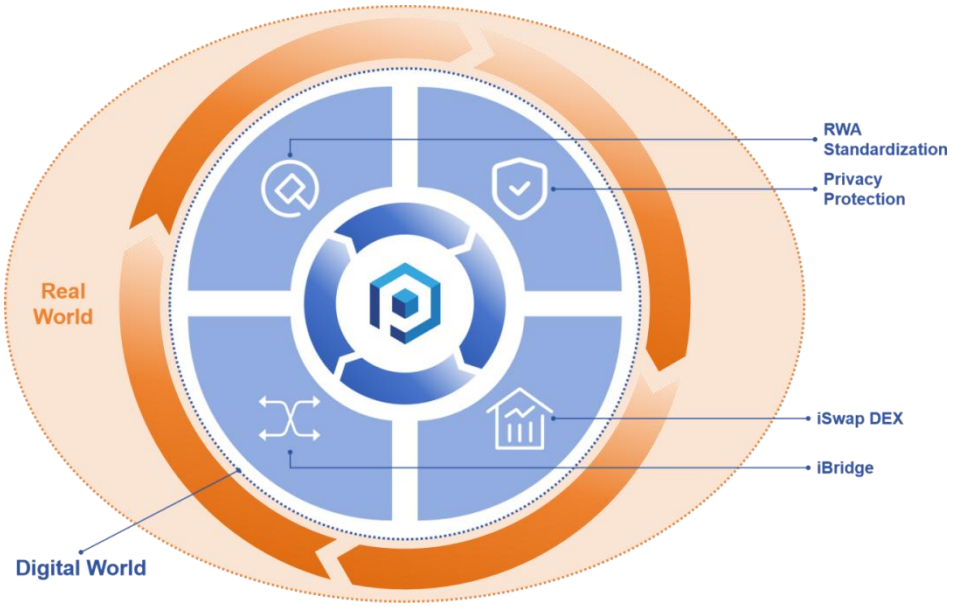
4.1.1 Four Pillars of Decentralized Digital Asset Infrastructure

- **RWA (Real-World Assets) Standardization Platform**: In the traditional financial system, the digitization of physical assets faces many challenges, such as inconsistent asset definitions and lack of standardized processes. The platform uses advanced blockchain technology and smart contracts to accurately digitize and standardize various physical assets. From real estate, intellectual property, commodities to corporate accounts receivable, any physical asset with economic value can be digitally mapped on this platform. This standardized processing not only lays a solid foundation for subsequent asset transactions and circulation, but also enables different types of physical assets to interact efficiently in the same ecosystem, greatly improving the liquidity and market value of assets.
- **iSwap DEX**: iSwap DEX plays a core trading hub role in the ecosystem. Due to its decentralized nature based on blockchain, it completely eliminates the involvement of intermediaries, which means that users can trade digital assets directly on the blockchain without trusting third parties. Users only need to initiate transaction instructions in their digital wallets, and the smart contract will automatically match and complete the delivery of assets

according to the rules. At the same time, since iSwap runs on the blockchain, all transaction records are public and cannot be tampered with, which provides regulators and market participants with a clear traceability path for transaction data, further enhancing the market's trust.

- **iBridge multi-chain cross-chain bridge**: Portaldot Network ecosystem has successfully broken down the barriers between multiple blockchains by introducing cross-chain bridge technology, and achieved seamless connection with other blockchain networks. iBridge cross-chain bridge can safely and efficiently realize the transfer and interaction of digital assets between different blockchains, including mainstream blockchains such as Bitcoin and Ethereum, which enables various digital assets in the ecosystem to circulate in a wider range of blockchain networks, greatly expanding the application scenarios and market scope of assets.
- **Privacy protection suite based on ZKP**: In order to provide users with strong privacy protection capabilities, we will develop a privacy protection suite based on ZKP (Zero-Knowledge Proof) to encrypt user sensitive data while ensuring that their transactions are legal and valid, and ensuring the fairness and security of transactions.

4.1.2 Dual Circulation Between "Digital World" and "Physical World"



In the digital world of the Portaldot Network ecosystem, standardized digital assets continue to increase in value through circulation and circulation in the ecosystem. When digital asset tokens are traded on DEX, their prices fluctuate based on market supply and demand, changes in the value of the asset itself, and the development of the entire ecosystem. As more and more investors participate in digital asset trading, the market activity continues to increase, and the value of digital assets continues to increase. In addition, we also encourage developers to develop innovative applications. These applications can provide more usage scenarios and value-added services for digital assets, further promote the value growth of digital assets, enable digital assets to play a role in more complex financial scenarios, and achieve higher value returns. This continuous appreciation of digital assets in the digital world has attracted more physical asset owners to digitize their assets and add them to the ecosystem, forming a virtuous circle.

The added value of the digital world has a positive feedback effect on the physical world. As the value of digital assets continues to increase, the circulation of corresponding physical assets in the physical world has also been promoted. At the same time, since the price fluctuations of digital asset tokens reflect the market's expectations of physical assets, this also provides important market signals for physical asset owners, prompting them to allocate and manage their physical assets more reasonably.

The dual circulation of the digital world and the physical world also strengthens the financial attributes of physical assets. Through digital assetization, physical assets can be more conveniently used for financial operations such as mortgages and guarantees, further expanding the financing capabilities and financial application scenarios of physical assets. This strengthening of financial attributes not only helps to improve the utilization efficiency of physical assets, but also injects new vitality into the innovation and development of the entire financial system.

Under the interrelated circulation of the digital world and the physical world, the wealth value of the Portaldot Network ecosystem continues to grow. On the one hand, the appreciation of digital assets enables investors and asset owners to obtain more economic returns, which can be further invested in the trading of digital assets and the development of physical assets, forming a new growth momentum. On the other hand, the circulation and development of physical assets have driven the prosperity of the real economy and created more employment opportunities and social wealth. This growth in wealth value not only benefits individuals and enterprises participating in the ecosystem, but also makes a

positive contribution to the economic development of the entire society.

4.2 Community Driven Platform: Organizational Paradigm of Web3.0

In the ecosystem of Portaldot Network, the community platform occupies a pivotal position. It is jointly built by the volunteer community and the technical community to build an open and collaborative ecological governance system, which continuously injects vitality and vigor into the ecology and promotes our continuous innovation and development.

As an important cornerstone of the ecosystem, the volunteer community plays an irreplaceable role in the construction and promotion of the ecology. They are active in various fields, from the promotion of projects to the service support of users, from the organization and planning of community activities to the dissemination of ecological culture, they are everywhere, laying a solid mass foundation for the development and growth of the ecosystem.

The technical community is the core driving force of the Portaldot Network ecosystem. It consists of three parts: the technical platform, the technical incentive platform, and the hackathon competition platform:

- As the core component of the technology community, the technology platform is further divided into the open source community and the developer platform. The open source community brings together the world's top blockchain developers and technology enthusiasts, who share their codes, technical insights and innovative ideas here to promote the continuous evolution of blockchain technology. The developer platform provides developers with a series of convenient development tools and services to help developers build applications based on Portaldot Network more efficiently.
- The technology incentive platform is another major feature of the technology community. It encourages developers to actively participate in the technological innovation and ecological construction of the ecosystem by establishing a complete incentive mechanism. The technology incentive platform provides developers with a variety of incentive methods, including but not limited to digital currency rewards, project equity distribution, and medal of honor awards. When developers achieve outstanding results in technological innovation, application development, community contribution, etc., the technology incentive platform will

give corresponding rewards.

- The hackathon platform is an innovation incubator for the technology community. We will hold hackathons regularly, inviting developers, designers, entrepreneurs, etc. from all over the world to gather together to develop innovative projects around specific themes within a specified time. In the hackathon, participants can give full play to their imagination and creativity to develop various innovative and practical applications and solutions. Through the hackathon, we can discover and tap into innovative projects with potential and incorporate them into the ecosystem for incubation and cultivation.

The volunteer community and the technical community do not exist in isolation in the Portaldot Network ecosystem, but rather they work in close collaboration with each other. The volunteer community has created a good atmosphere and environment for the development of the technical community through publicity and promotion, user services, etc., attracting more technical talents to pay attention to and participate in the ecosystem. The technical community, through continuous technological innovation and application development, has provided the volunteer community with more activity content and service carriers, enhancing the influence and value of the volunteer community.

4.3 Cutting Edge Integration Platform: Advancing Technological Frontiers

As a key component of the Portaldot Network ecosystem, the cutting-edge technology integration platform shoulders the important mission of leading technological innovation and promoting the implementation of applications. The platform not only focuses on technology research and development, but also focuses on transforming advanced technologies into practical applications to create real value for users.

Under the current trend of technological development, AI, metaverse and quantum computing have become the most promising and influential fields. Integrating blockchain with them is expected to produce synergistic effects, break through the limitations of traditional technologies, and open up new application scenarios and business models. By integrating resources from all parties, the platform brings together the world's top technical talents and scientific research institutions to form an open and innovative ecological environment to promote exchanges and cooperation between different

technologies.

4.3.1 Large Scale and Multimodal Model Research

We have invested a lot of resources and energy in the research of big models and multimodal big models. On the one hand, we are committed to optimizing the architecture and training algorithms of existing big models to improve the efficiency and performance of the models. On the other hand, we focus on the integration and understanding of multimodal data, which can effectively integrate and analyze data of different modalities. This cross-modal interaction method enables the model to better understand the semantic relationship behind the data and improve the execution effect of multimodal tasks.

In addition, we are also actively exploring the application of big models in the field of blockchain. For example, the language processing capabilities of big models can be used to automatically audit and detect vulnerabilities in smart contracts on blockchains. With the help of multimodal big models, the visualization and intelligent interaction of blockchain data can be realized, and the operating status and value of blockchains can be more intuitively understood.

4.3.2 Privacy Computing Research

When we studied privacy computing, we chose homomorphic encryption and federated computing collaboration. Homomorphic encryption is a special form of encryption that allows specific operations to be performed on ciphertext, and the result is the same as the result of performing the corresponding operation on the plaintext and then encrypting it. In the field of AI, homomorphic encryption can be used to train and reason models on encrypted data without decrypting the data, thereby protecting data privacy. Federated computing is a distributed computing framework that enables joint data analysis and model training by establishing a secure data collaboration mechanism between multiple participants.

4.3.3 Metaverse Research

Portaldot Network actively explores the application of blockchain technology in the metaverse, and is committed to creating an open, secure, and sustainable metaverse ecosystem. First, we use the decentralized nature of blockchain to build the infrastructure of the metaverse, manage user identities, assets, and transactions in the virtual world, and ensure the authenticity and immutability of data. Secondly, we want to provide a fair and transparent creation and revenue platform for creators and

developers in the metaverse. Through smart contracts, creators can digitally confirm the ownership of their works and set reasonable revenue rules. This mechanism can encourage more creators to participate in the content creation of the metaverse and enrich the ecology of the metaverse.

In the future, we will cooperate with technical teams such as VR, AR, and brain-computer research to promote the development of immersive experience technology for the Metaverse. By combining the advantages of blockchain and advanced interactive technology, we will create a more realistic and smooth Metaverse experience for users.

4.3.4 Quantum Computing Research

Recognizing the possible impact of quantum computing on blockchain technology, we are actively conducting research on the integration of blockchain and quantum cryptographic computing. Since quantum computing may pose a threat to existing blockchain encryption algorithms, we are committed to researching new encryption algorithms that are resistant to quantum computing attacks to ensure the security of blockchain systems, and have achieved some globally renowned results in this field. Of course, quantum computing also brings new opportunities to blockchain. The powerful computing power of quantum computing can accelerate the transaction processing speed of blockchain and improve the throughput of the system. At the same time, the use of quantum algorithms can optimize the consensus mechanism of blockchain, reduce energy consumption, and improve the efficiency of the network. In addition, we will also explore the value of quantum computing in blockchain data analysis and mining. Quantum computing has unique advantages in processing large-scale data, which can help blockchain systems better understand and utilize data and discover potential value and laws.